



MALLA REDDY TECHNICAL CAMPUS
A Constituent Unit of MALLA REDDY VISHWAVIDYAPEETH
Deemed to be University
Approved by UGC & AICTE New Delhi
Maisammaguda, Medchal (Dist), Hyderabad - 500100, Telangana



EXAM ROOM ALLOCATION SYSTEM

PROJECT INTRODUCTION

One-line idea :

An automated system that efficiently allocates students to examination rooms based on available room capacity.

Team Members :

A.HARINI

M.DEEKSHA

K.RAKSHITHA

Problem Domain:

Education Technology (EdTech) / Examination Management

THE CHALLENGE

Real-world problem :

- Manual room allocation takes a lot of time.
- Managing many students and rooms is difficult.

Why it matters :

- Saves time for faculty.
- Reduces human errors.
- Makes examination management easier.
- Provides organized room and seating information.

Key challenges faced :

- Handling large number of students
- Managing different room capacities
- Avoiding duplicate student allocation
- Preventing incorrect room assignments
- Reducing manual processing time

OUR INNOVATIONS

Core idea

Automatically allocate students to examination rooms based on available room capacity and student details.

Key features

- Automatic room allocation
- Capacity-based assignment
- Duplicate-free allocation
- Room-wise student list
- Fast and organized output

Unique approach

Algorithm-based allocation instead of manual room assignment.

HOW IT WORKS

Main flowchart :

INPUT

Student Details + Room Details



PROCESSING

Validate Data & Check Capacity



ALGORITHM

Allocate Students to Rooms



OUTPUT

Room-wise Student Allocation

User journey :

Admin → Enter Data → Run Allocation → View Result

TECHNOLOGY & ALGORITHM

Tech stack :

Frontend: HTML, CSS

Backend: python

Main algorithm

Capacity-Based Room Allocation Algorithm

Why chosen ?

- Simple to implement
- Efficient for student allocation
- Prevents room overcapacity
- Easy to scale

Complexity

Time Complexity: $O(n)$

FUTURE VISION

Current achievements

- Automated room allocation
- Capacity-based assignment
- Organized room-wise output

Future improvements

- Automatic seat-number allocation
- QR-based verification
- PDF/Excel report generation

Scalability

- Multiple departments & examination
- Multiple buildings/campuses

Final takeaway

“Smart allocation for smoother and more efficient examinations.”

THANK YOU !